

AION Clinical Time-Series — Evaluation Rubric

Reference: PhysioNet PTB Diagnostic Database & CinC Challenge 2019 Sepsis Cohort

1. ECG Track — 12-Lead Myocardial Infarction Detection

Submission format: per-patient CSV with columns patient_id, predicted_class, confidence, top_3_leads_affected.

Metric	Weight	Target	Notes
Per-lead macro F1 (12 standard leads)	40%	≥ 0.90	Class-imbalance aware; equal weight per lead.
Sensitivity at specificity ≥ 0.90	25%	≥ 0.85	Clinical operating point; missed MI is the costliest failure.
ST-elevation localization (II/III/aVF/V5/V6)	20%	≥ 0.95	Infero-lateral MI: 2 inferior + 2 lateral leads must agree.
Inference budget (CPU, single 5s strip)	10%	≤ 30 s	Clinical realism; disqualifying if exceeded.
Interpretability (SHAP lead attribution)	5%	≥ 0.80	Attribution heatmap must highlight II/III/aVF/V5/V6 for MI.

Public benchmark anchor: PhysioNet PTB Diagnostic Database (<https://physionet.org/content/ptbdb/1.0.0/>). 290 patients, 52 diagnostic classes. Top SOTA F1 for MI detection: ~ 0.92 - 0.95 . Common traps: lead mis-ordering, gain misapplication, test-set leakage via patient stratification.

2. ICU Track — Sepsis Onset Prediction (2h/4h/6h horizons)

Submission format: per-minute CSV with columns timestamp_iso, sepsis_prob, horizon_h.

Metric	Weight	Target	Notes
AUROC at 6h horizon	35%	≥ 0.85	Primary lead metric; horizon-aware training required.
Sensitivity at specificity 0.85 (6h)	20%	≥ 0.45	Clinically-actionable operating point.
AUROC at 2h and 4h horizons	15%	$\geq 0.75 / 0.80$	Progressive difficulty; reports both numbers.
Expected Calibration Error (ECE)	15%	< 0.05	False alarms erode clinician trust.
Per-hour precision-recall curve area	10%	> 0.30	Hour-stratified; accounts for regime switches.
Interpretability (IG / SHAP attributions)	5%	≥ 0.75	Top features should include MAP, lactate slope, SpO2.

Public benchmark anchor: PhysioNet CinC Challenge 2019

(<https://physionet.org/content/challenge-2019/1.0.0/>). 40,336 ICU patients, ~7% sepsis prevalence. Top team AUROC ~0.85, sensitivity@0.85-spec ~0.45-0.50. Common traps: lookahead bias in horizon training, sparse-lactate handling, class imbalance.

3. Combined Gating & Safety Constraints

Both tracks must pass the following gates or the submission is rejected:

Gate	Threshold	Failure consequence
Reproducibility (fixed seed, single CPU)	3 runs within 1% absolute	Automatic re-train with stronger seed control.
Provenance disclosure	All data sources named, including synthetic data is not flagged.	Rejection from challenge.
Method family coverage	≥ 3 stat, ≥ 3 ML, ≥ 3 DL, ≥ 3 T	80% coverage is met.
Visual semantic loop	Reference PNGs analyzed, insights documented	Rejection if analysis is attached to report.
Failure-case analysis	At least 1 false-negative case discussed	Block until at least 1 case is documented.
Governance approval	ts-critic + c-critic both sign off	Cannot stop before approval.

4. Submission Format Details

ECG track — per-patient prediction CSV:

```
patient_id, predicted_class, confidence, top_3_leads_affected
```

Example rows:

```
P001, myocardial_infarction_inferolateral, 0.93, II;aVF;V5
P019, myocardial_infarction_inferolateral, 0.89, III;aVF;V6
P136, healthy_control, 0.97, (empty)
```

ICU track — per-minute prediction CSV:

```
timestamp_iso, sepsis_prob, horizon_h
```

Example rows (hours 0-15 of the 48h sepsis patient):

```
2024-01-15 10:00:00, 0.08, 6
2024-01-15 10:01:00, 0.09, 6
2024-01-15 13:30:00, 0.62, 4
2024-01-15 16:00:00, 0.81, 2
```

5. Data Provenance

ECG: PhysioNet PTB Diagnostic Database, 3 patients sampled (P001, P019, P136). WFDB .dat/.hea/.xyz files preserved at original fidelity in data/raw_external/ptb_ecg/patient*/. CSV export in long format for adapter compatibility.

ICU: PhysioNet CinC Challenge 2019 reference structure. MIMIC-III and eICU-CRD full distributions returned HTTP 403 during data preparation; a high-fidelity synthetic cohort was generated grounded in Sepsis-3 / Surviving Sepsis Campaign setpoints (HR 75 to 135, MAP 90 to 55, lactate 1.0 to 4.5, SpO2 99 to 88 with on-off oxygen titration). Synthetic provenance must remain visible in any final report.

6. Documented Failure Modes

The model must explicitly discuss at least one failure mode per track.

ECG:

- Lead mis-ordering (Frank vs Mason-Likar) causes features to be physically meaningless. Mitigation: explicit lead-ordering assertion in data/interface.py.
- Gain misapplication (200 vs 1000) inflates or deflates amplitudes. Mitigation: read gain from .hea header before scaling.
- Patient stratification leakage across train/val splits. Mitigation: stratified group K-fold by patient_id.

ICU:

- Lookahead bias: using future vitals to predict sepsis onset. Mitigation: rolling-window feature construction with explicit cutoff timestamps.
- Sparse-lactate handling: lactate is sampled every 2-4h; forward-fill can leak. Mitigation: forward-fill with mask; treat NaN as feature, not bug.
- Class imbalance: ~7% sepsis prevalence. Mitigation: class-weighted loss + threshold tuning on validation set.

7. Interpretability Score

For both tracks, the interpretability score (weight 5%) is computed as:

$$\text{interpretability} = 0.5 * (\text{clinical_feature_recall}) + 0.3 * (\text{attribution_concentration}) + 0.2 * (\text{qualitative_agreement})$$

Where:

- `clinical_feature_recall` = fraction of top-10 SHAP/IG features that match a clinical-knowledge prior list.
- `attribution_concentration` = $1 - \text{normalized entropy of the attribution distribution}$ (penalizes diffuse, non-actionable explanations).
- `qualitative_agreement` = human reviewer score (1-5) on whether the attribution narrative is plausible.

ECG clinical priors: ST deviation in II/III/aVF/V5/V6, Q-wave depth, R-wave progression V1-V6, T-wave polarity in lateral leads.

ICU clinical priors: MAP drop, lactate rise, SpO2 desaturation, HR rise, temperature trajectory, fluid balance.